Preview

Correct answers are hidden.

Score for this quiz: 25 out of 25
Submitted Mar 9 at 8:09am
This attempt took 14 minutes.

Question 1
1 / 1 pts

Multiple Inheritance vs Implementations

Choose the correct statement(s) about Java.

- [ ] In Java, a class can extend more than one parent class (Multiple inheritance).
- [x] In Java, a class can implement more than one interface

Question 2
1 / 1 pts

Consider the following Java codes.

Choose the keywords that can be omitted without causing compilation failure.

```
public abstract interface Matter {              // A
  public final static int MAX_COUNT = 1234567;  // B
  public abstract double getMassKgs();          // C
}
```

- [x] abstract in A
- [x] public in B
- [x] final in B
- [x] static in B
- [x] public in C
- [x] abstract in C

Question 3
1 / 1 pts

Consider the following codes. Choose the boolean variables that have value true at the end of the program execution.

```
class MyInt {
    int i = 0;
    public MyInt (int x ) {  i= x;}
    public boolean equals( Object o ) { return i == ((MyInt)o).i; }
}
public class Overridden {
    public static void main(String[] args) {
        MyInt x1 = new MyInt (1);
        MyInt x2 = new MyInt (1);
        Object y1 = x1;
        Object y2 = x2;
        boolean a = x1.equals(x2);
        boolean b = y1.equals(y2);
        boolean c = x1 == x2;
        boolean d = y1 == y2;
    }
}
```

- [x] a
- [x] b
- [ ] c
- [ ] d

Question 4

2 / 2 pts

Invisible Constructors

There are 2 invisible constructor-related things that Java adds:

- **Where there is no ctor defined in a class, an invisible minimal no-args ctor will be added.**
- **Where a ctor does not start with "super...", an invisible 1st-line call to superclass no-args ctor will be added.**

Which of the following codes will compile correctly?

#1:

```
class A {}
public class B extends A {}
```

#2:

```
class A {
    public A (int x ) {};
}
public class B extends A {
}
```

#3:

```
class A {
    public A (int x ) {};
}

public class B extends A {
    public B (int x ) {};
}
```

#4:

```
class A {
    public A (int x ) {}
}
public class B extends A {
    public B (int x ) { super (x); }
}
```

- [x] #1
- [ ] #2
- [ ] #3
- [x] #4

Question 5

1 / 1 pts

Overriding vs Overload

Which of the following class definition will override the inherited toString method correctly and compile successfully?

class A {
  String toString () { return "AAA"; }
}

class B {
  public String toString () { return "BBB"; }
}
class C {
  Override
  String toString () { return "CCC"; }
}

class D {

```
  @Override
  public String toString () { return "DDD"; }
}


class E {

  @Override
  public String toString ( String msg ) { return msg ; }
}
```

- [ ] A
- [x] B
- [ ] C
- [x] D
- [ ] E

Question 6

1 / 1 pts

Consider the following Java codes. Determine the contents of A and B.

```
class Molecule {
   String whoAreYou() { return "I'm a molecule"; }
}
public class DNA extends Molecule {
   @Override
   String whoAreYou() { return "I'm some DNA"; }
   public static void main( String[] args ) {
       DNA covidEnvGene = new DNA();
       Molecule m = covidEnvGene;
       String A = covidEnvGene.whoAreYou();
       String B = m.whoAreYou();
       System.out.println ( A );
       System.out.println ( B );
   }
}
```

A

I'm some DNA

B

I'm some DNA

Question 7

3 / 3 pts

Polymorphism

Consider the following code snippet.

- Among statements labelled A,B,C,D,E, choose the one that will print out the different message from others.
  - A . (fill in a single uppercase letter A,B,C,D, or E)
- (y/n) The ctor of class Star is overloaded.
  - y (fill in either y or n)
- (y/n) if we remove the no-args ctor of class Star (with label X), the program still compile and run.
  - n

```
class Star {
  double mass;
  Star(double mass) { this.mass = mass; } ;
  Star () { };                      // X
}
class RedGiant extends Star {
    RedGiant() { super(); }
    @Override                 // ask Compiler to check if the following does override ...
    public String toString(){//overriding the toString() method defined in Object
        return "RedGiant with mass "  + mass ;
    }
}
public class Cast {
    public static void main(String[] args) {
        Star x = new Star ();
        RedGiant y = new RedGiant();;
        Star z = y;
        System.out.println (x);  // A
        System.out.println (y);  // B
        System.out.println (z);  // C
        System.out.println (y.toString() );  // D
        System.out.println (z.toString() );  // E
    }
}
```

Answer 1:

A

Answer 2:

y

Answer 3:

n

Question 8

1 / 1 pts

throws FileNotFoundException

Consider the following code snippet:

```
public static void main(String[] args) throws FileNotFoundException
```

Which of the following statements about this code is correct?

- The main method is designed to catch and handle all types of exceptions.
- The main method will not terminate if any exception occurs.
- The main method is designed to catch and handle the `FileNotFoundException`.
- The main method terminates if the `FileNotFoundException` occurs.

Question 9

1 / 1 pts

Handling file exceptions:

- Your program must read in an existing text file.
- You want the program to terminate if any exception related to the file occurs.

Which of the following indicates the correct code for the header of the `main` method?

- `public static void main(String[] args) throws FileNotFoundException`
- `public static void main(String[] args)`
- `public static void main(String[] args) throws IOException`
- `public static void main(String[] args) throws UnknownFileException`

Question 10

1 / 1 pts

Consider the following code snippet.

```
PrintWriter outputFile = new PrintWriter("payrollReport.txt");
```

Which of the following statements about the `PrintWriter` object is correct?

If a file named `"payrollReport.txt"` already exists, a new file named `"payrollReport_1.txt"` will be created and used.

If a file named `"payrollReport.txt"` already exists, an exception will occur.

If a file named `"payrollReport.txt"` already exists, existing data will be deleted before new data are added to the file.

If a file named `"payrollReport.txt"` already exists, new data will be added to the end of the file.

Question 11

1 / 1 pts

Insert the missing code in the following code fragment.

This fragment is intended to

- read an input file named `dataIn.txt` and
- write to an output file named `dataOut.txt`.

```
public static void main(String[] args) throws FileNotFoundException
{
   String inputFileName = "dataIn.txt";
   String outputFileName = "dataOut.txt";
   File inputFile = _________________;
   Scanner in = new Scanner(inputFile);
   . . .
}
```

`new File(inputFile)`

`new File(System.in)`

`new File(outputFileName)`

`new File(inputFileName)`

Question 12

1 / 1 pts

Creating a File object:

Which of the following is the correct syntax for creating a `File` object?

`File inFile = new File("input.txt");`

`File inFile = File.open("input.txt");`

`File inFile = File("input.txt");`

`File inFile = new File.open("input.txt");`

Question 13

1 / 1 pts

Override equal()

Consider the following codes. Choose the boolean variables that have value true at the end of the program execution.

```
class MyInt {
    int i = 0;
    public MyInt (int x ) {  i= x;}
    public boolean equals( MyInt o ) { return i == o.i; }            // A
    public boolean equals( Object o ) { return i == ((MyInt)o).i; } // B
}
public class Overridden {
   public static void main(String[] args) {
       MyInt x1 = new MyInt (1);
       MyInt x2 = new MyInt (1);
       Object y1 = x1;
       Object y2 = x2;
       boolean a = x1.equals(x2);
       boolean b = y1.equals(y2); // Should b be true if a is true?
       boolean c = x1 == x2;
       boolean d = y1 == y2;
   }
}
```

- [x] a
- [x] b
- [ ] c
- [ ] d

Question 14

1 / 1 pts

Override The equal() Method

Indicate the statements that have syntax error.

```
class MyInt {
    int i = 0;
    public MyInt (int x ) {  i= x;}
    @Override public boolean equals( MyInt o ) { return i == o.i; }              // A
    @Override public boolean equals( Object o ) { return i == ((MyInt)o).i; }    // B
}

Hints:
Does statement A override method equals defined in class Object?
Does statement B override method equals defined in class Object?
```

- (•) A
- ( ) B

- Both A and B
- Neither A nor B

Question 15

1 / 1 pts

- (y/n) In Java, a class *can extend more than one parent class (Multiple inheritance).*
  - n
- *(y/n) In Java, a class can implement more than one interface.*
  - y

Answer 1:

n

Answer 2:

y

Question 16

1 / 1 pts

Override vs Overload

Which of the following class definition will override the inherited toString method correctly and compile successfully?

```
class A {
   String toString () { return "AAA"; }
}

class B {
   public String toString () { return "BBB"; }
}
class C {
   String toString () { return "CCC"; }
}

class D {
   public String toString () { return "DDD"; }
}
class E {
   public String toString ( String msg ) { return msg ; }
}
```

- [ ] A
- [x] B
- [ ] C

- [x] D
- [ ] E

Question 17

1 / 1 pts

Invisible Constructors

There are 2 invisible ctor-related things that Java adds:

- **Where there is no ctor defined in a class, an invisible minimal no-args ctor will be added.**
- **Where a ctor does not start with "super...", an invisible 1st-line call to superclass no-args ctor will be added.**

Consider the code snippet below, identify the statement that will cause the compilation error.

```
class BlackHole {
  private String name;
  public BlackHole(double mass) { } ;       // ctor
}
public class SMBH extends BlackHole {
  public SMBH() { super(4000000.0);  }      // A
  public SMBH(int x) { super(x);  }         // B
  public SMBH(double x) { super (x); }      // C
  public SMBH() { }                         // D
}
```

- [ ] A
- [ ] B
- [ ] C
- [x] D

Question 18

2 / 2 pts

Consider the codes below. Choose the correct statements below.

```
class Star {
}
public class BlackHole extends Star {
    private String name;
    public int BlackHole(double mass) {  return 0; };     // A
    public static void main(String[] args) {
        BlackHole x = new BlackHole();
        int y = x.BlackHole(10.0);
    }
}
```

- [ ] ctor must have no return statement
- [x] ctor must have no return type

- [x] The codes compile and run
- [ ] The ctor of class BlackHole is overloaded
- [ ] Any non-constructor methods cannot have the same name as the class.
- [ ] Line A is a ctor.

Question 19

1 / 1 pts

Consider the following code snippet. Choose the statements that compile.

```
class Star {
  double mass;
  Star(double mass) { this.mass = mass; }
}
```

```
class RedGiant extends Star {
  RedGiant() { super(); } // A
  RedGiant(int mass) { super(mass); } // B
  RedGiant(double mass) { super(mass); } // C
}
```

- [ ] A
- [x] B
- [x] C
- [ ] None is a correct answer.

Question 20

1 / 1 pts

Boxing/Unboxing

Consider the following codes. Choose the boolean variables that have value true at the end of the program execution.

```
public class Overridden {
   public static void main(String[] args) {
       Integer x1 = 1;  //
       Integer x2 = 1;  // x1 and x2 are distinct object created by users.
       Object y1 = x1;
       Object y2 = x2;
       boolean a = x1 == x2;
       boolean b = x1.equals(x2);
       boolean c = x1 == x2;
       boolean d = y1 == y2;
        System.out.println (a);
        System.out.println (b);
        System.out.println (c);
        System.out.println (d);
   }
```

```
}
```

```
Hints: Java pre-build objects for (small) integers between -128 and 127 in the memory pool.
```

- [x] a
- [x] b
- [x] c
- [x] d

Question 21

1 / 1 pts

Boxing/Unboxing

Consider the following codes. Choose the boolean variables that have value true at the end of the program execution.

```
public class Overridden {
   public static void main(String[] args) {
       Integer x1 = 1000;  //
       Integer x2 = 1000;  // x1 and x2 are distinct object created by users.
       Object y1 = x1;
       Object y2 = x2;
       boolean a = x1 == x2;
       boolean b = x1.equals(x2);
       boolean c = x1 == x2;
       boolean d = y1 == y2;
        System.out.println (a);
        System.out.println (b);
        System.out.println (c);
        System.out.println (d);
   }
}
```

- [ ] a
- [x] b
- [ ] c
- [ ] d

Quiz Score: 25 out of 25